

PRAVEEN VIGHNESH S T

Mechatronics & robotics engineer who designs actuated robotic hardware and brings it up in ROS2 and Gazebo. Hands-on across the full stack: mechanical design, motion control, embedded integration, and real-time computer vision.

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SKILLS

Robotics	ROS2 (Jazzy), ros2_control, MoveIt2, RViz, Gazebo, Isaac Sim, OpenCV (Knowledge: URDF/Xacro, IK/FK, DH parameters, Joint state publisher, FSM, SLAM)
Electronics	KiCad, Altium Designer, Circuit Designer IDE, EasyEDA, Arduino IDE, Raspberry Pi 5 OS, UART (Perfboard circuit assembly, multi-rail power, RC filtering, decoupling capacitors, PLC & PID control)
CAD & Simulation	Fusion 360, SolidWorks, Siemens NX, CREO, CATIA V5, Ansys, MATLAB Simulink, Simscape (DFM/DFA, RCA, BOM creation, MBSE, FEA, Geometric Dimensioning & Tolerancing)
Fabrication	3D printing, pillar drill, grinding, laser cutting, Sheet metal bending, CNC, welding & soldering (Prusaslicer, cold pulling, overhang & support strategy, bend radius and relief, corner radius)
Programming & Tools	Python, C++, PyTorch, Claude Code, MCP for Fusion 360 & MS office, GIT

EDUCATION

M.Sc. Electromobility, FAU Erlangen, Germany	10/2023 – Present
B.Tech. Mechanical Engineering, VIT, Vellore	07/2018 – 06/2022

PRACTICAL EXPERIENCE

Factory Automation and Production Systems, Nürnberg

- Master Thesis** **03/2026 – Present**
- Building a custom **7-DOF** robotic system with **10kg** payload capacity achieving under **20 arcmin** backlash
 - Automated the ROS2 development workflow with **Claude Code**, programmatically generating Xacro, launch files, and repetitive package configuration, cutting **simulation setup time** and speeding up debugging process
 - Implemented CANopen CiA 402 motor control using interpolated (mode 7) and cyclic synchronous position (mode 8) modes for smooth, synchronized multi-axis motion
 - Simulated dynamic joint motion in **Gazebo** with **ros2_control**, assigning realistic link masses and inertias to **quantify peak torque** demand at each joint under load, directly driving the motor and bearing sizing calculation

- Project Thesis** **06/2025 – 01/2026**
- Designed and integrated custom mechatronic subsystems for an automated beverage-mixing machine, fabricating a **perfboard control circuit** that streamlined wiring and reduced testing and debugging time by **50%**
 - Reduced total project cost from 5000€ to 3000€ (**~40% savings**) by using off the shelf parts and designing modular, Lego-style components reducing 3D printing time from **13 to 6 hours**
 - Designed a grasshopper-linkage speed multiplier that raised actuator output speed from **30 to 72 mm/s** while sustaining the required **10 N** dispensing force

- ISOLogic Automation, Tiruppur – Design Engineer** **08/2022 – 12/2022**
- Collaborated with cross-functional teams to design and manufacture a 41-ft Pratt truss structure for a 40,000 sq. ft. industrial shed, through precision cutting, welding, machining, and post-processing operations

- Cognizant Technology Solutions, Chennai – Programmer Analyst Trainee** **02/2022 – 07/2022**
- Implemented automated testing with Selenium & TestNG to improve efficiency, increase test coverage and stability, and cut post-release defects by 15%

PROJECTS

- Vision-Based Foot Gesture HMI for a Medical Robot**
- Designed a hands-free HMI that translates foot gestures into real-time robot control commands using computer vision, implementing a finite state machine with temporal filtering, baseline calibration, and tracking-loss tolerance for reliable gesture recognition under fast motion and partial occlusion

- Anthropomorphic Robot Hand Design**
- Built a two four-bar linkage for fingers with an under-actuated mechanism which operates the MCP joint to move the PIP and DIP joint dependently, reproducing a human-like grasping trajectory prior to object contact

- Adaptive Gripper with EDS foam-based Sensing**
- Designed an adaptive parallel jaw gripper printed with TPU95A for flexibility and equipped with EDS foam for touch detection during compliant grasping

LANGUAGE & CERTIFICATIONS

Languages: German (B1) | English (C1) | Tamil (C2, native)

Certifications: Electric Vehicles & Mobility | Introduction to Self-Driving Cars | Engineering Design with Fusion 360